

Sunhwa Bang

Post-doc researcher at Gyeongsang National University, Research Institute of Molecular Alchemy

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EDUCATION

Ph.D.	2023	Earth and Environmental Science	Seoul National University
M.Sc.	2017	Earth and Environmental Science	Seoul National University
B.Sc.	2014	Earth System Sciences (double major)	Yonsei University
B.Eng	2014	Environmental Engineering (double major)	Yonsei University

RESEARCH INTERESTS

Isotope Geochemistry, Sedimentology, Chemostratigraphy, Paleoceanography, Paleoclimatology, Environmental Simulations

PUBLICATIONS

In English

1. [Bang, S.](#), & Lee, Y. I. (2020). Darriwilian carbon isotope stratigraphy in the Taebaeksan Basin, Korea and its implications for Middle Ordovician paleoceanography. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 541, 109534.
2. [Bang, S.](#), Huh, Y., Khim, B. K., Takata, H., Ikehara, M., Hyeong, K., ... & Cho, H. (2022). Deep-Water Circulation over the Last Two Glacial Cycles Reconstructed from Authigenic Neodymium Isotopes in the Equatorial Indian Ocean (Core HI1808-GPC04). *Ocean Science Journal*, 1-10.
3. Kwon, H., [Bang, S.](#), Joo, Y. J., & Sim, M. S. (2024). Local and global controls on carbon and sulfur isotope records in the Middle Ordovician Epeiric sea successions of the Taebaeksan Basin, Korea. *Chemical Geology*, 122216.
4. [Bang, S.](#), & Huh, Y. (2024). Reconstructing CO₂ uptake capacity and pH dynamics in the Middle Ordovician Taebaeksan Basin, Korea. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 655, 112520.
5. Jeong, J., [Bang, S.](#), Wilson, D. J., & Huh, Y. (2025). Middle Ordovician climate modulation by enhanced continental and reverse weathering: Evidence from lithium isotopes. *Chemical Geology*, 122859.
6. Kim, M., [Bang, S.](#), Ji, Y., & Kim, H. I. (2025). Utilization of natural opal-CT-bearing mudstone as a precursor of zeolite-geopolymer composites: Synthesis and characterization. *Case Studies in Construction Materials*, e05470.
7. [Bang, S.](#), Keum, J. Y., Ji, Y., Kang, Y. J., So, B. D., Hong, J. K., & Kim, H. I. (2026). Application of YOLOv11 deep learning model for classification and counting ice-rafted debris (IRD) in core sediments in the Arctic Ocean. *Artificial Intelligence in Geosciences*, 100191.

In Korean

1. S. Bang, Y. Lee (2016). A review of the middle Darriwilian carbon isotope excursion, Middle Ordovician. Journal of the Geological Society of Korea, v.52, no.6, p. 917-935 (in Korean with English abstract)

PROFESSIONAL SERVICES & ACTIVITIES

Research Institute of Molecular Alchemy, Gyeongsang National University,
Post-doctoral researcher, 2023-

IGC2024 organization committee, International Geological Congress
ESG TF team, 2023-2024

Center for deep-surface coupling of Earth, Seoul National University
Post-doctoral researcher, 2023

Korean carbonate rock geochemistry research team, Seoul National University
researcher, 2020-2023

R/V ISABU Indian Ocean research team, Seoul National University
MC-ICPMS analyst, 2018-2019
Onboard researcher, 2017

Northwest Pacific boron isotope research team, Seoul National University
researcher, 2017-2020

Teaching Assistant Graduate Fellowship, Seoul National University,
School of Earth and Environmental Sciences, 2017-2018

Ocean Mineral Research Center, Korea Institute of Ocean Science & Technology
Research assistant, 2016-2017

Brain Korea 21 Research Fellowship, Seoul National University, 2014-2015

PROFESSIONAL MEMBERSHIPS

The Geological Society of Korea, since 2015

American Geophysical Union, since 2018

Geochemical Society, since 2025

Korean Society of Oceanography, since 2025

HONORS & AWARDS

Postdoc LAMP Fellowship, Gyeongsang National University, Research Institute of Molecular Alchemy, 2023-

Brain Korea 21 Outstanding Graduate Students Rewards, Seoul National University, 2021

Teaching Assistant Graduate Fellowship, Seoul National University, School of Earth and Environmental Sciences, 2017-2018

Brain Korea 21 Research Fellowship, Seoul National University, 2014-2015

SKILL SET

Expedition.	R/V & Field research, Sampling (sediment core, rock, river, microfossils)
Analysis	MC-ICPMS, QQQ-ICPMS, IC, XRD, SEM, Microscopes
Software	R, Ocean Data View
Certificates	Engineer of Soil Environment (Since 2012) K-IODP Core School (2015, 2018) J-DESC Core School Basic Course (2016) SNU Static Research Software R Course (2020)